On the Maximal Cross-Correlation of Algebraically Constructed Costas Arrays

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***Abstract*—Families of Costas arrays with low pairwise cross-correlation are sought. The two families of all exponential Welch arrays and all Golomb arrays generated in a certain finite field are specifically studied, and the maximal cross-correlation is determined by exhaustive search. Mathematically rigorous explanations for some of the observed results are presented, a surprising link between Welch and Golomb arrays is revealed, and what remains to be proved is stated precisely. The results suggest that the families with uniformly low cross-correlation correspond to finite fields whose size is a safe prime power.**

***Index Terms*—Costas arrays, cross-correlation, Golomb method, safe primes, Welch method.**

I. Introduction

COSTAS arrays are square arrays of dots/1's and blanks/0's, such that there exists exactly one dot per row and column (that is, they are permutation arrays), and such that 1) no four dots not lying on a straight line form a parallelogram, 2) no four dots lying on a straight line form two equidistant pairs, and 3) no three dots lying on a straight line are equidistant. They appeared for the first time in 1965 in the context of SONAR detection ([2], and later [3] as a journal publication), when Costas, disappointed by the poor performance of SONARs, used them to describe a novel frequency hopping pattern for SONAR systems with optimal autocorrelation properties (namely, autocorrelation sidelobes of height at most 1). At that stage their study was entirely empirical and application-oriented. In 1984, however, after the publication of the two main construction methods for Costas arrays by Golomb [9], still the only ones of general applicability available today, they officially acquired their present name and they became an object of mathematical interest and study.

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While Costas arrays can indeed be defined by their ideal "thumbtack" autocorrelation property (along with the fact that they are permutation arrays), little is known mathematically about their cross-correlation. Perhaps the first result relevant to the cross-correlation of Costas arrays was the demonstration by Freedman and Levanon [8] in 1985 that any two Costas arrays of the same order (trivially small orders excepted) must have a maximal cross-correlation of at least 2, and hence that ideal cross-correlation and ideal autocorrelation are incompatible properties. The study of cross-correlations became more focused in later years, with the work by Chang and Scarbrough in 1989 [1], which demonstrated that there exist sets of Welch Costas arrays whose pairwise cross-correlation is ideal, provided that only a strip of the correlation surface around the origin is considered, and not the entire surface. Later, in 1994, Maric, Seskar, and Titlebaum [12] studied the cross-correlation of Welch Costas arrays more thoroughly, providing a table with the maximal observed cross-correlation between any two Welch Costas arrays of order $p-1, p$ prime (effectively a first version of our Table I below), and showing that this cross-correlation is bounded above by $\frac{p-1}{2}$ (a first version of our Theorem 1 below). The conference version of this work had appeared in ICASSP 1990 [15], where the authors had already observed that some primes corresponded to local minima in the table, and had identified them to be safe primes.

An important part of the puzzle was still missing, though, namely cross-correlation results for Golomb–Costas arrays. The subject had been considered by Drumheller and Titlebaum in 1991 [7], who suggested an upper bound that is, unfortunately, not always tight. Then, Rickard, in his Master's thesis in 1993 [13], extended the cross-correlation tables for Welch arrays of [15] and offered the corresponding results for Golomb arrays. He observed a surprising link between the Welch and Golomb results (given the two construction methods are so dissimilar), confirmed the special role of the safe primes for the Golomb results as well, and asked whether this link could be explained, though he did not offer any explanation. Our present work effectively picks up from that point, providing new and more detailed rigorous explanations than previously available.

Why would one be interested in the cross-correlation of Costas arrays? On the one hand, as we are about to see, the cross-correlation of Costas arrays presents a rich mathematical behavior, so it is worth studying as a mathematical subject in its own sake. On the other hand, as also described in [7], [12], there are indeed applications where low cross-correlation is important, maybe as important as low auto-correlation. More specifically, consider two or more SONAR systems operating

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TABLE I

VALUE OF THE MAXIMAL CROSS-CORRELATION BETWEEN PAIRS OF W_1 ARRAYS AND OF G_2 ARRAYS GENERATED IN $\mathbb{F}(p)$, FOR THE FIRST FEW PRIMES $p > 4$: ENTRIES IN RED CORRESPOND TO SAFE PRIMES, WHILE 19 APPEARS IN BLUE BECAUSE OF A SLIGHT IRREGULARITY IN THE G_2 CASE. W_1 RESULTS FOR $p < 80$ FIRST APPEARED IN [12], [15]; W_1 RESULTS FOR $p < 200$ AND G_2 RESULTS FOR $p < 80$ FIRST APPEARED IN [13]

Prime	W_1	G_2	Prime	W_1	G_2	Prime	W_1	G_2	Prime	W_1	G_2
5	2	2	59	5	12	127	42	41	197	98	97
7	2	2	61	30	29	131	26	25	199	66	65
11	3	4	67	22	21	136	68	67	211	70	69
13	6	5	71	14	13	139	46	45	223	74	73
17	8	7	73	36	35	149	74	73	227	6	13
19	6	6	79	26	25	151	50	49	229	114	113
23	4	6	83	5	9	157	78	77	233	116	115
29	14	13	89	44	43	163	54	53	239	34	33
31	10	9	97	48	47	167	6	12	241	120	119
37	18	17	101	50	49	173	86	85	251	50	49
41	20	19	103	34	33	179	6	12	257	128	127
43	14	13	107	5	10	181	90	89	263	7	12
47	5	8	109	54	53	191	38	37	269	134	133
53	26	25	113	56	55	193	96	95	271	90	89

in the same sea region, each configured with its own (Costas) frequency hopping pattern. The operation of one of them is bound to cause interference to the others, and therefore we would like to shield SONAR systems from that type of multiuser cross-interference. Although the cross-correlation cannot be optimally small (namely at most 1), as we saw above, it may still be possible, for special families of Costas arrays, for it to be much smaller than the order of the arrays used, in which case the cross-interference can still be kept to a level acceptably low for applications.

In order to convey to the reader not only the results of our research, but also the motivation that led to their discoveries, we deliberately adopt an informal style of discussion in the presentation of our results, as opposed to the "orthodox" theorem-proof structure favored by purists, without, of course, sacrificing expositional rigor. In particular, our work progresses from observation to theoretical explanation, and from the general to the more specific. The advantage of this approach is that it reveals to the reader the difficulties and impasses which force us to gradually restrict the scope of our investigation.

II. WHY COSTAS ARRAYS? SOME MOTIVATION

RADAR and SONAR systems detect the distance and velocity of targets around them by periodically transmitting a waveform W and listening for reflections R. Assuming an ideal noiseless environment, R is just a copy of W, only attenuated and shifted in frequency and time. The time delay indicates the distance of the target, while the frequency shift, through the Doppler effect, its velocity (we assume here that the frequency content of W is narrowband enough for the Doppler effect, which is multiplicative, to be well approximated by a uniform additive shift for all frequencies).

The simplest way to detect the time and frequency shifts is by applying a matched filter: R is cross-correlated with shifted versions of W for various time and frequency shifts, and the pair of shifts corresponding to the maximal cross-correlation are the true shifts sought:

$$(s_t^*, s_f^*) = \underset{(s_t, s_f)}{\operatorname{argmax}} |\psi(s_t, s_f)|$$
$$\text{where } \psi(s_t, s_f) = \int_{-\infty}^{+\infty} (F_{s_f} W)(t - s_t) R(t) dt. \quad (1)$$

Here, s_t and s_f denote the time and frequency shifts, respectively, and the operator F_{s_f} maps W to the signal obtained by translating every positive frequency of W's spectrum by s_f to the right and every negative frequency by s_f to the left (so that $F_{s_f} W$ remains real).

Alas, this simple idea fails to work in practice, because all real media are incoherent: phase delay varies with frequency, hence waveforms tend to spread while traveling in the medium [3], so that, by the time R reaches the receiver, it may look so different from W that coherent processing becomes inappropriate.

Costas's idea [3] was to discard the unreliable phase information, and carry out the cross-correlation based on the energy contents of W and R alone. Consider a waveform as follows:

$$W(t) = A \cos\left(\phi_k + 2\pi \left(f_0 + \frac{a_k}{n} f_1\right) t\right)$$
$$t \in \left[\frac{k-1}{n} T, \frac{k}{n} T\right] \quad (2)$$

where $k \in [n]$, T is the time duration of the pulse, f_0, f_1 are two predetermined frequencies, ϕ_k are phases suitably chosen so that the phase of W is continuous in t (we may choose $\phi_1 = 0$), and $a : [n] \to [n]$ is a bijection. This is a frequency hopping waveform whose instantaneous frequency is

$$f(t) = f_0 + \frac{a_k}{n} f_1, t \in \left(\frac{k-1}{n} T, \frac{k}{n} T\right), k \in [n]. \quad (3)$$

We observe that W is completely determined by a, given that f_0, f_1, and T are set.

Costas's idea amounts effectively to placing an energy content detector before the matched filter, thus reconstructing a from W, and similarly for R: the signals fed to the matched filter can then each be abstracted as a 2D infinite sequence, representing the time-frequency plane. This sequence is full of 0's/blanks (energy is not present), except for a $n \times n$ square that corresponds to a permutation array (whose 1s/dots denote that energy is present there). The filter overlays the two 2D sequences, then shifts one with respect to the other by v rows vertically and u columns horizontally, and counts how many pairs of dots overlap: this is the value of the cross-correlation Ψ at (u, v) (compare with Definition 1 below). In the absence of noise, R

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is an exact copy of W, only shifted in time and frequency, so the matched filter will have found the correct shift parameters when the cross-correlation becomes equal to n.

When noise is present, however, some of R's dots may have shifted irregularly, gone altogether missing, or even new dots may have appeared: R will no longer be an exact copy of W, and the maximal cross-correlation will no longer be n. One will have no alternative than to design the filter to locate the maximal cross-correlation (whose value will no longer be known a priori) and return the shift parameters corresponding to it. This maximum may no longer be unique and/or one of the (former) sidelobes may have grown taller than the main lobe. Unfortunately, both cases result in spurious target detection.

What should the form of a (or the corresponding array A) be in order to minimize the probability of spurious detections? In the absence of noise, the cross-correlation is just a shifted form of the autocorrelation of A, so we need to choose A in such a way as to suppress the height of the autocorrelation sidelobes relatively to the main lobe of height n as much as possible:

$$A = A^* = \arg\min_{A} \max_{(u,v)} \Psi_{A,A}(u,v).$$

Choosing any pair of dots in A, there exists a shift (their distance vector) moving one on top of the other, so sidelobes of height 1 will exist no matter what. If, however, we stipulate that distance vectors be unique, there will be no sidelobe of height 2 or more. This, however, is precisely the Costas property (compare with Definition 4 below)! Autocorrelation is known as *autoambiguity* in the SONAR/RADAR community, and waveforms with the Costas property are said to have *ideal thumbtack autoambiguity* [3], [10].

Why should the optimal A be a permutation array, as we assumed (summarily and without any further explanation) above? Would using twice the same frequency, or using two frequencies simultaneously, not improve the autocorrelation? Costas argued on basic engineering principles that indeed it would not [3].

Why, finally, do we need the full force of the Costas property? Would sidelobes of height 2 or more, but still much shorter than the main lobe, not be enough? Indeed they might not, but for a reason we have not mentioned so far: even if such a signal performed well under noise conditions, as discussed above, there is still the issue of multipath interference. In the real world R will most likely be the sum of attenuated and shifted noisy versions of W (by different attenuation and shift parameters) representing echoes from different reflection paths, which may add up constructively at the receiver: the taller the sidelobes of W are, the easier it is for them to add up to a significantly tall sidelobe in R.

III. Basics

In this section, we give precise definitions for all the terms used in the Introduction, as well as for everything else needed. Throughout this work, we will be using the notation $[n]$ to denote the set of the first n positive integers $\{1,\dots,n\}, n \in \mathbb{N}$. It will also be convenient to consider primes to be included in prime powers as a subset. Whenever we want to explicitly exclude them, we will be referring to *strict* prime powers.

A. Definition of the Costas Property

Simply put, a Costas array is a square arrangement of dots and blanks, such that there is exactly one dot per row and column, and such that all vectors between dots are distinct.

Definition 1: Let $f : [n] \to [n]$ be a bijection. Then, f is said to have the *Costas property* iff the collection of vectors $\{(i-j, f(i)-f(j)) : 1 \le j < i \le n\}$, called *the distance vectors*, are all distinct, or, equivalently, iff

$$\begin{aligned} &\forall i,j,k \text{ such that } 1 \le i,j,i+k,j+k \le n : \\ &f(i+k)-f(i) = f(j+k)-f(j) \Rightarrow i = j \text{ or } k = 0 \end{aligned} \tag{4}$$

in which case f is called a *Costas permutation*. The corresponding *Costas array* A_f is the square $n \times n$ array where the elements at $(f(i), i), i \in [n]$ are equal to 1 (dots), while the remaining elements are equal to 0 (blanks):

$$A_f = [a_{ij}] = \begin{cases} 1, & \text{if } i = f(j) \\ 0, & \text{otherwise} \end{cases}, \quad j \in [n]. \tag{5}$$

Remark 1: The operations of horizontal flip, vertical flip, and transposition on a Costas array result in a Costas array as well: hence, out of a Costas array eight can be created, or four if the particular Costas array is symmetric.

B. Construction Algorithms

There are two main algorithms for the construction of Costas arrays, based on the algebraic theory of finite fields. We state them below without proofs (which can be found in [4], [9] in full detail). The numerical indices used to label them signify how much smaller the order of the resulting Costas arrays are than the size of the finite field used. Both methods can actually be extended to yield a wide range of submethods [4], [10], which, however, will not concern us further here.

Definition 2: (Exponential Welch Construction $W_1(p,\alpha,c)$): Let p be a prime, α a primitive root of the finite field $\mathbb{F}(p)$, and $c \in [p-1]-1$. The *exponential Welch permutation* corresponding to α and c is defined by

$$f(i) = \alpha^{i-1+c} \bmod p, i \in [p-1]. \tag{6}$$

Remark 2: Given a W_1 permutation, it is well known that its horizontal and vertical flips also correspond to W_1 permutations. Its transpose, however, does not: it is what we define as a *logarithmic Welch permutation*. The distinction is well defined as, for $p > 5$, there are no symmetric W_1 arrays [5]. We will no further consider logarithmic Welch permutations in this work, so "Welch" will henceforth be synonymous with "exponential Welch."

Definition 3: (Golomb Construction $G_2(p,m,\alpha,\beta)$): Let $q = p^m$, where p prime and $m \in \mathbb{N}^*$, and let α, β be primitive roots of the finite field $\mathbb{F}(q)$. The Golomb permutation corresponding to α and β is is defined through the equation

$$\alpha^i + \beta^{f(i)} = 1, i \in [q-2]. \tag{7}$$

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Remark 3: The horizontal and vertical flips of a G_2 permutation are themselves G_2 permutations, just like in the Welch case. This time, however, the same holds true for transpositions as well.

C. Cross-Correlation

We now give a precise definition of the cross-correlation between two Costas arrays:

Definition 4: Let $f, g : [n] \to [n]$ where $n \in \mathbb{N}^*$, and let $u, v \in \mathbb{Z}$. The cross-correlation between f and g at (u, v) is defined as

$$\Psi_{f,g}(u,v) = \left|\{(f(i)+v, i+u) : i \in [n]\} \cap \{(g(i), i) : i \in [n]\}\right|. \quad (8)$$

Informally, we can think of the cross-correlation in the following way. First, we place the two Costas arrays on top of each other, and then we translate the first by v units vertically and u horizontally. The number of pairs of overlapping dots is the value of the cross-correlation Ψ at (u, v).

Remark 4: Neither of the two "+" signs in (8) should be interpreted as a modular addition: they both represent addition in $\mathbb{Z}$. Doing so, however, may lead to more tractable equations which provide useful bounds for the cross-correlation (as we shall see below).

D. Safe Prime Powers

The following family of primes, which has played an important role in the proof of Fermat's Last Theorem [14], will also appear in the results presented later in this work. We will also need the extension of this family to prime powers:

Definition 5: A prime power q is *safe* iff $q = 2r+1$, where r is prime. For q prime, the corresponding r is known as a *Sophie Germain prime*.

Strict safe prime powers have a very specific form. Indeed, let $q = p^n, n \in \mathbb{N}^*, p$ prime, be the strict prime power considered: we would like $p^n - 1 = 2r, r$ odd prime, but $p^n - 1$ is clearly divisible by $p^m - 1$ as long as $m|n$. Therefore, either $p^m - 1 = r$ or $p^m - 1 = 2$. The former leads to $p = 2$ since q is odd; the latter leads to $p = 3, m = 1$. Since $2^n = 1 + 2r$ is impossible, we conclude that $p = 3$. We also conclude that n has to be a prime, otherwise $3^m - 1, m|n, 1 < m < n$ is a nontrivial even divisor of $3^n - 1$. We have proved:

Lemma 1: Strict safe prime powers are of the form $3^p, p$ prime.

IV. Cross-Correlation Observations

Our ultimate goal is to find families of Costas arrays as large as possible (of the same order), in which the cross-correlation of any pair and for any shift is uniformly bounded by a number much smaller than the order. Perhaps the two most natural families to consider are the family of all W_1 arrays in $\mathbb{F}(p), p$ prime, and of all G_2 arrays in $\mathbb{F}(q), q$ prime power. In the case of W_1 arrays, however, there is a fine point we need to pay attention to: cross-correlating a W_1 array with a cyclic shift of itself in $\mathbb{F}(p)$ can lead to any value between $p-2$ and 1, hence such cross-correlations have to be excluded.

Definition 6: For f, g W_1 arrays in $\mathbb{F}(p)$, the notation $f \overset{\text{pr}}{\neq} g$ will denote not only that f and g are not the same, but also that they are not cyclic shifts of each other.

We computed exhaustively $\max_{(u,v)} \max_{(f_1 \overset{\text{pr}}{\neq} f_2)} \Psi_{f_1,f_2}(u,v)$ and $\max_{(u,v)} \max_{(g_1 \neq g_2)} \Psi_{g_1,g_2}(u,v)$ where f_1, f_2 range over all W_1 arrays in $\mathbb{F}(p)$ and similarly g_1, g_2 range over all G_2 arrays in $\mathbb{F}(p), 5 \leq p \leq 271$. The computational time needed is not long: in fact, all computations presented in this work can be comfortably carried out on a single CPU in a day's work using Matlab. The results are shown in Table I in the columns labeled W_1 and G_2, respectively, which is an adapted and extended version of the tables found in [12], [13]. We make the following three observations:

- In the black rows, columns W_1 and G_2 are related in a straightforward way: $G_2 = W_1 - 1$. There is also a somewhat more elusive relation between columns W_1 and Prime. Consider, for example, $p = 43$, and factorize $p-1$ to get $42 = 2 \cdot 3 \cdot 7$, writing the factors from the lowest to the highest. Find the first factor following the initial factor of 2, and divide $p-1$ by it: $42/3 = 14$. The result is the corresponding entry in column W_1! As another example, consider $p = 181$: $p - 1 = 180 = 2^2 \cdot 3 \cdot 5$, hence the first factor following the initial factor of 2 is also a 2. It follows that $180/2 = 90$, which is indeed the corresponding entry in column W_1. It appears then that $W_1 = (p-1)/t$, where t is the smallest prime dividing $(p-1)/2$.
- In the red rows, our observations regarding the black rows no longer hold true, and there appears to be no obvious relation between the columns. There are still, however, two observations we can make. First of all, these appear to be the primes we are interested in, as the entries in both W_1 and G_2 seem to correspond to local minima: for example, the W_1 entries for $p = 257$ and 269 are 128 and 134, respectively, while $p = 263$ which lies in between and corresponds to a red row leads to a W_1 entry of only 7. The situation for the G_2 entries is similar. But then, what are these primes? If we try, for example, the factorization trick on $p = 179$, we get $178 = 2 \cdot 89$, and 89 is a prime. Similarly, $107 = 1 + 2 \cdot 53$, and 53 is a prime. So, these primes appear to be of the form $p = 1 + 2q$, where q is a prime; but these are precisely the safe primes, as defined in Definition 5.
- The unique blue row in the table corresponds to $p = 19$, marking it as a semi-irregular case: the W_1 entry is correctly 6 as $19 - 1 = 2 \cdot 3^2$ and $18/3 = 6$, but G_2 is also 6 as opposed to $W_1 - 1 = 5$. Is 19 an one-off exception or are there other primes like it? We do not know for sure, though Lemma 2 below shows that 19 is the only "quadratically safe" prime!

Though it is tempting to also deduce from Table I that the entries of the W_1 column over the red rows seem to be nondecreasing, this turned out actually not to be true: the maximal cross-correlation for the next two safe primes $p = 347$ and 359

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TABLE II
VALUE OF THE MAXIMAL CROSS-CORRELATION BETWEEN PAIRS OF G_2 ARRAYS GENERATED IN $\mathbb{F}(q)$, FOR THE FIRST FEW STRICT PRIME POWERS q: ENTRIES IN RED CORRESPOND TO IRREGULAR CASES, WHILE 16 APPEARS IN BLUE BECAUSE IT SEEMS TO BE SEMI-IRREGULAR

Prime power	G_2	Prime power	G_2	Prime power	G_2
4	1	32	6	128	9
8	3	49	23	169	83
9	3	64	20	243	21
16	5	81	39	256	84
25	11	121	59	289	143
27	6	125	61	343	113

is found to be 6, which is less than the value of 7 corresponding to $p = 263$.

Lemma 2: The equation $p = 2r^2 + 1, p$ and r prime, has $p = 19, r = 3$ as its only solution.

Proof: The possible values of $r \bmod 3$ are 0, 1, and -1; the first case leads to $r = 3, p = 19$, while the last two can be combined into $r^2 \equiv 1 \bmod 3$, hence $2r^2 + 1 \equiv 0 \bmod 3$, implying that $3|2r^2 + 1$, hence p cannot be prime. ∎

We also computed exhaustively $\max_{(u,v)} \max_{(g_1 \neq g_2)} \Psi_{g_1,g_2}(u,v)$ where g_1, g_2 range over all G_2 arrays in $\mathbb{F}(q), q$ a strict power of a prime, $4 \leq q \leq 256$. The results are shown in Table II. As before, we can make some observations:

- In the black rows corresponding to odd powers, we can apply the same rule as before: $G_2 = (q-1)/r - 1$, where r is the smallest prime dividing $(q-1)/2$. For even powers, namely powers of 2, we need a small modification: $G_2 = (q-1)/r - 1$, where r is the smallest prime dividing $(q-1)$. For example, $64 - 1 = 63 = 3^2 \cdot 7$, and $63/3 - 1 = 20$, while $256 - 1 = 255 = 3 \cdot 5 \cdot 17$, and $255/3 - 1 = 84$.
- In the red rows, there appears to be again no obvious relation between the columns, but, once more, the corresponding prime powers appear to be the ones we are interested in, because the corresponding G_2 entries seem to correspond to local minima. These prime powers seem to fall into two categories: powers of 2 of the form 2^k such that $2^k - 1$ is a Mersenne prime, hence unfactorizable (such is the case for, e.g., $q = 32$ and 128), and odd safe prime powers, when $q - 1$ is factorizable as 2 times a prime. The only representative of the latter category in the table is $27 = 1 + 2 \cdot 13$.
- The unique blue row corresponds to $q = 16$ and the situation here is an exact parallel of $p = 19$ in Table I: $16 - 1 = 15 = 3 \cdot 5$, and $15/3 - 1 = 4$, while the corresponding G_2 entry is actually 5, so higher by 1 than it should be, just as in the case for $p = 19$ in Table I.

In the next two sections, we will attempt to prove rigorously some of the observations catalogued here.

V. CROSS-CORRELATION BETWEEN W_1 ARRAYS

The cross-correlation between exponential Welch arrays at the origin is quite simple to compute, and has the potential to lead to much more interesting and general results. The calculation is shown below in several steps.

A. The General Case and Some Simplifications

Consider two W_1 arrays f_1 and f_2, generated by α_1 and c_1, and by $\alpha_2 \neq \alpha_1$ and c_2, respectively (remember that we do not consider the cross-correlation of a W_1 array with a cyclic shift of itself, hence we do not allow $\alpha_1 = \alpha_2$). What is their cross-correlation at (u,v)? We need to solve the system

$$i_2 = i_1 + u, \alpha_1^{i_1 - 1 + c_1} \bmod p + v = \alpha_2^{i_2 - 1 + c_2} \bmod p. \quad (9)$$

Note the "hybrid" addition used here (and below): representatives of the residue classes $\bmod p$ in the range $\{0, \ldots, p-1\}$ are considered to be ordinary integers, and then added to v using ordinary integer addition. As a result, the system is, at first sight, intractable. Since, however

$$i_2 = i_1 + u \Rightarrow i_2 \equiv i_1 + u \bmod (p-1) \quad (10)$$

this system leads to the equation (this being an one-way step):

$$\alpha_1^{i_1 - 1 + c_1} \bmod p + v = \alpha_2^{i_1 + u - 1 + c_2} \bmod p. \quad (11)$$

Since both α_1 and α_2 are primitive roots, $\exists r \in [p-1] - 1 : (r, p-1) = 1, \alpha_2 = \alpha_1^r, r \neq 1$, hence

$$\alpha_1^{i_1 - 1 + c_1} \bmod p + v = \alpha_1^{r(i_1 + u - 1 + c_2)} \bmod p \quad (12)$$

and, dropping the subscript in α_1 for simplicity, as it is no longer needed:

$$\alpha^{i - 1 + c_1} \bmod p + v = \alpha^{r(i + u - 1 + c_2)} \bmod p. \quad (13)$$

A further simplification that still causes no loss of generality can be introduced by observing that the term $u + c_2$ in the formula above allows for the same range of values as c_2 alone. We can then focus on the case $u = 0$:

$$\alpha^{i - 1 + c_1} \bmod p + v = \alpha^{r(i - 1 + c_2)} \bmod p. \quad (14)$$

B. A Further Simplification and a Bound

Unfortunately, (14) seems still to be intractable in its full generality, so we attempt a new simplification. Taking modulo p throughout, we get

$$\alpha^{i - 1 + c_1} + v \equiv \alpha^{r(i - 1 + c_2)} \bmod p. \quad (15)$$

Note that this is again an one-way step, not an equivalence, unless $v = 0$. Setting $\alpha^{i-1} = x$, we obtain the polynomial equation

$$x^r \alpha^{rc_2} - x\alpha^{c_1} - v \equiv 0 \bmod p. \quad (16)$$

It is clear that any solution of (14) is also a solution of (16), so, defining the sets

$$S(v, c_1, c_2, \alpha, r) = \{x \in \mathbb{F}^*(p) : x^r \alpha^{rc_2} - x\alpha^{c_1} - v \equiv 0 \bmod p\} \quad (17)$$

and

$$T(v, c_1, c_2, \alpha, r) = \left\{x \in \mathbb{F}^*(p) : \alpha^{i - 1 + c_1} \bmod p + v = \alpha^{r(i - 1 + c_2)} \bmod p\right\} \quad (18)$$

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we have demonstrated that

$$\Psi_{f_1,f_2}(u,v) \le |T(v,c_1,c_2,\alpha,r)| \le |S(v,c_1,c_2,\alpha,r)| \tag{19}$$

and that

$$\Psi_{f_1,f_2}(u,0) = |T(0,c_1,c_2,\alpha,r)| = |S(0,c_1,c_2,\alpha,r)|. \tag{20}$$

In other words, the cross-correlation is bounded above by the number of roots of the polynomial in (16), and is therefore at most r; a similar derivation appears in [7], where the authors reach the same conclusion.

C. Cross-Correlation at the Origin

At this point, we revisit (14) in order to study the only case which seems tractable, namely where $u = v = 0$. This substitution yields

$$\begin{aligned} &\alpha^{i-1+c_1} \bmod p = \alpha^{r(i-1+c_2)} \bmod p \Leftrightarrow \\ &\quad i-1+c_1 \equiv r(i-1+c_2) \bmod (p-1) \\ &\quad \Leftrightarrow (r-1)(i-1+c_2) \equiv c_1 - c_2 \bmod (p-1) \end{aligned} \tag{21}$$

(this equation also appears in [7]). Setting $s := (r-1, p-1)$, we know by the theory of linear congruences (see [11], p. 51) that this equation has s solutions if $s|(c_1 - c_2)$, and no solution otherwise. Without loss of generality, then, we can choose $c_1 = c_2$, and seek solutions to the equation

$$(r-1)x \equiv 0 \bmod (p-1). \tag{22}$$

Clearly, the solutions form a subgroup of the group of integers modulo $p-1$, which confirms the fact that the number of solutions can only be a divisor of $p-1$.

How many can the solutions of (22) be when $r \neq 1$? The maximal number of solutions possible is $s = \frac{p-1}{2}$, which occurs iff $r - 1 = \frac{p-1}{2} \rightarrow r = \frac{p+1}{2}$, while at the same time $(\frac{p+1}{2}, p-1) = 1$ holds, so that this value of r is actually allowed. But, $1 = (\frac{p+1}{2}, p-1) = (\frac{p+1}{2}, 2) \rightarrow \frac{p+1}{2} \equiv 1 \bmod 2 \rightarrow p \equiv 1 \bmod 4$. It follows that (22) has $\frac{p-1}{2}$ solutions iff $p \equiv 1 \bmod 4$.

What if $p \equiv 3 \bmod 4$? Then we can aspire for at most $s = \frac{p-1}{3}$ solutions, which can occur only if $p \equiv 1 \bmod 3$ as well, so that this number be actually an integer. Assuming then that both of these congruences hold true, which is equivalent to $p \equiv 7 \bmod 12$, we check as before whether $(\frac{p-1}{3} + 1, p-1) = 1$. Setting $p = 7+12k$, we find this to be $(3+4k, 6+12k) = (3+4k, 3) = (4k, 3) = (k, 3)$, which is 3 if $3|k$ and 1, otherwise.

What happens if $3|k$? First of all, we can set $k = 3l$ and $p = 7 + 36l \rightarrow p \equiv 7 \bmod 36$. Let us take for example $p = 43$, in which case $\frac{p-1}{3} + 1 = 15$: choosing $r = 15$, it is true that (22) has $r - 1 = \frac{p-1}{3}$ solutions, except that we had no right to choose $r = 15$ in the first place, since $(15, 42) = 3$, as we just derived! What if we choose $r = 2 \cdot 14 + 1 = 29$? Then $(29, 42) = 1$, and $(28, 42) = (28, 14) = 14$. So, we can circumvent the difficulty choosing $r = 2\frac{p-1}{3} + 1$. Does this work in the general case? Indeed, for $p = 7 + 36l$ we get $(2\frac{p-1}{3} + 1, p-1) = (5 + 24l, 6 + 36l) = (5 + 24l, 1 + 6l) = (1, 1 + 6l) = 1$.

The next possible number of solutions of (22) is $\frac{p-1}{4}$. But this cannot possibly occur for any prime p: for, in order for this to be an integer, we need that $p \equiv 1 \bmod 4$, and we already saw that these primes actually lead to twice as many solutions!

Let us formulate a general argument now:[1] assume that $p \equiv 3 \bmod 4$, let w be a prime such that $p \equiv 1 \bmod w$, and let $r = \lambda\frac{p-1}{w} + 1$, for some $\lambda \in [w-1]$. It is clear that

$$\begin{aligned} s = (r-1, p-1) &= \left(\lambda\frac{p-1}{w}, p-1\right) \\ &= \left(\lambda\frac{p-1}{w}, w\frac{p-1}{w}\right) = \frac{p-1}{w}(\lambda, w) = \frac{p-1}{w}. \end{aligned} \tag{23}$$

We will show that r is admissible for either $\lambda = 1$ or $\lambda = 2$. Let $p = 1 + wk$, and observe that $(\lambda\frac{p-1}{w} + 1, p-1) = (\lambda k + 1, wk)$; unless $w|\lambda k + 1$, this is equal to $(\lambda k + 1, k) = (1, k) = 1$. Assuming now that $w|k+1$ and $w|2k+1$, it follows that $w|k$, so that $w|1$, a contradiction; so, either $\lambda = 1$ or $\lambda = 2$ leads to an admissible value of r. When is $\lambda = 2$ needed? When $w|k+1, k = lw - 1$ for some l, so that $p = 1 + (lw-1)w = lw^2 - w + 1 \leftrightarrow p \equiv w^2 - w + 1 \bmod w^2$.

We have shown that for every prime $p \equiv 3 \bmod 4$ and every prime w such that $p \equiv 1 \bmod (2w)$ (note that necessarily $w > 2$ so that $p-1$ is divisible by both 2 and w), there exists an admissible r such that (22) has $\frac{p-1}{w}$ roots. This is clearly maximized when w is the least possible such prime. For the same reason, it makes no sense to consider nonprime w (as in the case of $w = 4$ above), because a smaller prime divisor always exists in this case.

We can collect all of the above in the following theorem.

Theorem 1: Let f_1 and f_2 be W_1 permutations generated in $\mathbb{F}(p), p$ prime, and let r be the smallest prime such that $p \equiv 1 \bmod (2r)$. Then, $\max\limits_{\substack{\text{pr} \\ (f_1 \neq f_2)}} \Psi_{f_1,f_2}(u,0) = \dfrac{p-1}{r}$.

Remark 5: This result is a much more refined version of Theorem 4 in [12], which, in the notation of Theorem 1, proves that $\max\limits_{\substack{\text{pr} \\ (f_1 \neq f_2)}} \Psi_{f_1,f_2}(u,0) \le \dfrac{p-1}{2}$.

VI. Cross-Correlation Between Golomb Arrays

A. The General Case

Consider the two Golomb arrays generated in $\mathbb{F}(q), q$ power of a prime, by the primitive roots a, b, and α^r, β^s, respectively, where r and s are relatively prime to $q-1$. The generating equations are

$$\alpha^{i_1} + \beta^{j_1} = 1, \alpha^{ri_2} + \beta^{sj_2} = 1, i_1, i_2, j_1, j_2 \in [q-2]. \tag{24}$$

The cross-correlation between the two arrays at the point (u, v) is the number of solutions of the system

$$\alpha^{i_1} + \beta^{j_1} = 1, \alpha^{ri_2} + \beta^{sj_2} = 1, i_2 = i_1 + u, j_2 = j_1 + v. \tag{25}$$

It is important to note that the last two equations above hold in $\mathbb{Z}$, not modulo $q-1$, while the first two hold in the finite field $\mathbb{F}(q)$.

[1] In this instance of induction we move from a specific example to a general result, contrary to the general principle stated in the Introduction!

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B. A Simplification

System (25), when its last two equations are considered to hold modulo $q-1$, becomes:

$$\alpha^{i_1}+\beta^{j_1}=1, \alpha^{r(i_1+u)}+\beta^{s(j_1+v)}=1. \tag{26}$$

Solving the first equation for α^{i_1} and substituting into the second yields:

$$\alpha^{ru}(1-x)^r+\beta^{sv}x^s-1=0 \text{ with } \beta^{i_1}=x. \tag{27}$$

As every solution of (25) is also a solution of (26), but not necessarily vice versa, the value of the cross-correlation at (u,v) is bounded above by the number of solutions of the polynomial in (27). The two systems are, in general, completely equivalent iff $u=v=0$.

Theorem 2: Let f_1 and f_2 be the two Golomb permutations generated in $\mathbb{F}(q), q$ power of a prime, by the primitive roots α,β, and α^r,β^s, respectively, where r and s are less than and relatively prime to $q-1$. Then

$$\Psi_{f_1,f_2}(u,v) \le \left|\{x\in\mathbb{F}(q): \alpha^{ru}(1-x)^r+\beta^{sv}x^s-1=0\}\right| \tag{28}$$

where the inequality becomes an equality if $u=v=0$.

Again, a similar conclusion appears in [7], but considerably less refined.

C. Cross-Correlation at the Origin

The cross-correlation of G_2 arrays at the origin is substantially richer than that of W_1 arrays, and in a special case they are intimately related. Substituting $u=v=0$ in (28) leads to the equation

$$(1-x)^r+x^s-1=0. \tag{29}$$

This equation appears to be still beyond our reach, so we further restrict our scope to $s=1$. We then get:

$$\begin{aligned}(1-x)^r=1-x &\Leftrightarrow (1-x)^{r-1}=1\\ &\Leftrightarrow (r-1)y\equiv 0 \bmod (q-1)\\ &\quad \text{with } 1-x=\alpha^y, (r,q-1)=1.\end{aligned} \tag{30}$$

Note that $x\neq 0 \Leftrightarrow y\neq 0$ because of (27). In the special case when q is a prime, this is the same equation as in the W_1 case. More generally, when q is an odd prime power, the derivation preceding Theorem 1 applies verbatim, and, therefore, the number of solutions is the same. One of the solutions has to be rejected, however, namely $y=0\Leftrightarrow x=0$.

A novel case here, not considered in the derivation preceding Theorem 1, is q even, namely values of $q=2^n$ for some $n\in\mathbb{N}^*$: in these cases $q-1$ is either a (Mersenne) prime or else it does not contain factors of 2 because it is odd. Letting, as before, $w>2$ be a prime such that $q\equiv 1 \bmod w$, and setting $r=\lambda\frac{q-1}{w}+1, \lambda\in[w-1]$, it follows that the number of (admissible) roots of (30) is $(r-1,q-1)-1=(\lambda\frac{q-1}{w},q-1)-1=\frac{q-1}{w}(\lambda,w)-1=\frac{q-1}{w}-1$. It can be shown, in exactly the same way as in Theorem 1, that either $\lambda=1$ or $\lambda=2$ always leads to an admissible value of r.

Hence, the maximal number of solutions (30) can have corresponds to the least possible prime w. In particular, whenever $q-1$ is a Mersenne prime, (30) has the unique solution $y=0$ for any $r\neq 1$ which is not admissible, hence the cross-correlation at the origin in this case is equal to 0. We have proved the following theorem.

Theorem 3: Let f_1 and f_2 be W_1 permutations generated in $\mathbb{F}(p), p$ prime, and let f_1' and f_2' be G_2 permutations generated in the same field by some primitive roots α and β, and α^r and β, respectively, where $(r,p-1)=1, r>1$. Then

$$\max_{(f_1\neq f_2)}^{\text{pr}} \Psi_{f_1,f_2}(0,0)=\max_{(f_1'\neq f_2')}\Psi_{f_1',f_2'}(0,0)+1. \tag{31}$$

More generally, let f_1' and f_2' be G_2 permutations generated in the field $\mathbb{F}(q), q$ prime power, by some primitive roots α and β, and α^r and β, respectively, where $(r,q-1)=1, r>1$. Then

$$\max_{(f_1'\neq f_2')}\Psi_{f_1',f_2'}(0,0)=\frac{q-1}{t}-1 \tag{32}$$

where t is the smallest prime such that $q\equiv 1 \bmod (2t), q$ odd, or that $q\equiv 1 \bmod t, q$ even.

VII. Conjectures and Future Directions

The numerical evidence on the maximal cross-correlation between pairs of W_1 arrays and of G_2 arrays presented in Tables I and II, along with the theorems proved in the previous sections, together suggest some conjectures which, if true, greatly strengthen and broaden the results presented.

Letting f_1 and f_2 (f_1' and f_2') be W_1 (G_2) permutations generated in $\mathbb{F}(p)$ ($\mathbb{F}(q)$), p prime (q prime power), we introduce the shorthand

$$\Psi_{W_1}(p)=\max_{(u,v)}\max_{(f_1\neq f_2)}^{\text{pr}}\Psi_{f_1,f_2}(u,v),$$
$$\Psi_{G_2}(q)=\max_{(u,v)}\max_{(f_1'\neq f_2')}\Psi_{f_1',f_2'}(u,v). \tag{33}$$

A. Initial Conjectures

The following two conjectures are suggested only by the data presented in Tables I and II.

Conjecture 1: (Main Conjecture for Primes): With respect to $\Psi_{W_1}(p)$ and $\Psi_{G_2}(p)$, primes p can be classified into three groups:

- For non-safe primes $p\neq 19$ (such that $\frac{p-1}{2}$ is not a prime, that is), $\Psi_{W_1}(p)=\Psi_{G_2}(p)+1=\frac{p-1}{t}$, where t is the smallest prime such that $p\equiv 1 \bmod (2t)$.
- $p=19$ is the only prime for which $\Psi_{W_1}(p)=\Psi_{G_2}(p)=\frac{p-1}{t}=6=\frac{19-1}{3}$, 3 being the smallest prime t such that $19\equiv 1 \bmod (2t)$.
- Primes p such that $\frac{p-1}{2}$ is also a prime (safe primes, that is) correspond to local minima of both $C_{W_1}(p)$ and $C_{G_2}(p)$.

Observe that this conjecture suggests that the maximal cross-correlations of the two families of G_2 and W_1 arrays are closely related (they almost always differ by 1), despite the two families being of completely different nature. This is indeed remarkable!

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Conjecture 2: (Main Conjecture for Prime Powers): With respect to $\Psi_{G_2}(q)$, prime powers q can be classified into three groups:

- For prime powers $q \neq 16$ such that neither $\frac{q-1}{2}$ nor $q-1$ is a prime, $\Psi_{W_1}(q) = \frac{q-1}{t} - 1$, where t is either the smallest prime such that $q \equiv 1 \bmod (2t)$, if q is odd, or such that $q \equiv 1 \bmod t$, if q is even.
- $q = 16$ is the only prime power for which $\Psi_{G_2}(q) = \frac{q-1}{t} = 5$, where 3 is the smallest prime t such that $16 \equiv 1 \bmod t$.
- Prime powers q such that either $\frac{q-1}{2}$ is prime (in which case $q = 3^m$ for some prime m, as shown in Lemma 1) or $q-1$ is prime (in which case $q = 2^m$ for some prime m and $q-1$ is a Mersenne prime) correspond to local minima of $\Psi_{G_2}(q)$.

B. Revision

The theorems of the previous sections make some steps towards the proof of the two conjectures above. We can now be more specific regarding what remains to be shown as follows.

Conjecture 3: Assuming p is a non-safe prime, and that f_1 and f_2 are W_1 permutations generated in $\mathbb{F}(p)$, then $\Psi_{W_1}(p) = \max_{\substack{\text{pr} \\ (f_1 \neq f_2)}} \Psi_{f_1,f_2}(0,0)$. Furthermore, assuming $q \neq 16, 19$ is a non-safe prime power such that $q-1$ is not prime, $\Psi_{G_2}(q) = \Psi_{f_1',f_2'}(0,0)$, where f_1' and f_2' are G_2 permutations generated in $\mathbb{F}(q)$, such that, assuming the former is generated by primitive roots α and β, the latter is generated by α^r and β, where r is determined in the proof of Theorem 3.

This conjecture is immediately verifiable on Tables I and II, as the values of the W_1 column correspond precisely to the values obtained by Theorem 1, and the values of the G_2 column correspond precisely to the values obtained by Theorem 3.

Ironically, in accordance with the motivation for this work, as stated in the Introduction, safe prime powers, whose full behavior remains unaccounted for, are precisely the prime powers we are interested in the most, because, for both W_1 and G_2 arrays, the cross-correlation exhibits local minima there. What has been partly explained is the cross-correlation at the origin: if $p = 2q+1$ is a safe prime power, the maximum number of solutions of (22) and (30) will occur for $r = \frac{p-1}{q} + 1$, and it will be $r - 1 = 2$. Numerical evidence in Tables I and II clearly shows, however, that the cross-correlation maximum is higher than that, implying either that the maximum occurs away of (0, 0), or, as an additional alternative for the case of G_2 arrays, that it does occur at (0, 0), but for a different pair of permutations than the one stated in Conjecture 3.

C. Future Directions

It is clear that Conjecture 3 sets forth three tasks. First, we need to understand the results pertaining to the exceptional values of primes p and prime powers q: at the moment, our only observation is that the maximal cross-correlation there appears to be lower than elsewhere, but we are unable to qualify this statement further.

Second, we need to determine whether, for non-exceptional values of q, the maximal cross-correlation at (0, 0) over the family of G_2-permutations $G_2(q)$ occurs for the pair of permutations mentioned in the conjecture. To this end, it is enough to consider the polynomials derived in (29), and show that

$$P_{r,s}(x) = (1-x)^r + x^s - 1 \tag{34}$$

has at most as many roots as $P_{r,1}(x)$.

Third, we need to establish that, for nonexceptional values of primes p and prime powers q, the maximal cross-correlation values $\Psi_{W_1}(p)$ and $\Psi_{G_2}(q)$ correspond to cross-correlation values at (0, 0). No tangible progress has been made towards this direction, chiefly because of the "unalgebraic" form of the equations involved.

D. An Additional Partial Result

There has been some progress towards the second point of Section VII-C, in the form of a nontrivial upper bound for the number of roots of the polynomials $P_{r,r}$ (the special case where $r = s$) in $\mathbb{F}(q), q$ odd prime power, defined in (34). The result is based on a counting argument exploiting orbits of roots obtainable through symmetry transformations, which are applicable in $P_{r,r}, 1 < r < q-1, (r, q-1) = 1$ (which, in particular, implies that r is odd), but, unfortunately, not in the general polynomial $P_{r,s}$.

First, note that due to symmetry

$$P_{r,r}(x) = 0 \Leftrightarrow P_{r,r}(1-x) = 0. \tag{35}$$

Furthermore, assuming $x \neq 0$ (note that $x = 0$ is always a root)

$$\begin{aligned} P_{r,r}(x^{-1}) &= (1-x^{-1})^r + x^{-r} - 1 = x^{-r}[(x-1)^r + 1 - x^r] \\ &= -x^{-r}[(1-x)^r - 1 + x^r] = -x^{-r} P_{r,r}(x) = 0. \end{aligned} \tag{36}$$

Note that the fact that r is odd has been used. These two equations imply that roots come in orbits of at most six roots: assuming $x \neq 0, 1$ is a root, so are $1-x, x^{-1}, \frac{1}{1-x}, 1-x^{-1} = \frac{x-1}{x}$, and $\frac{x}{x-1}$. Roots $\{0, 1\}$ form an orbit of their own.

Subsequently, let $P_{r,r}(x) = P_{r,r}(y) = 0$: then $\frac{x}{y}$ is a root iff $\frac{1-x}{1-y}$ is a root (we assume $x, y \neq 0, 1$). This is because

$$\begin{aligned} (1-x)^r + x^r = 1 = (1-y)^r + y^r \Rightarrow \\ x^r - y^r = (1-y)^r - (1-x)^r \end{aligned} \tag{37}$$

and it follows that

$$\begin{aligned} \left(\frac{x}{y}\right)^r + \left(1 - \frac{x}{y}\right)^r = 1 &\Leftrightarrow x^r + (y-x)^r = y^r \\ &\Leftrightarrow x^r - y^r = (x-y)^r = (1-y)^r - (1-x)^r \\ &\Leftrightarrow \left(\frac{1-x}{1-y}\right)^r + \left(1 - \frac{1-x}{1-y}\right)^r = 1. \end{aligned} \tag{38}$$

Since, however, x^{-1} and y^{-1} are also roots, the above derivation yields that $\frac{x^{-1}}{y^{-1}} = \frac{y}{x}$ is a root iff $\frac{1-x^{-1}}{1-y^{-1}} = \frac{y}{x}\frac{1-x}{1-y}$ is a root. But $\frac{y}{x}$ is a root iff $\frac{x}{y}$ is a root, hence we conclude that

$$\begin{aligned} P_{r,r}\left(\frac{x}{y}\right) = 0 \Leftrightarrow P_{r,r}\left(\frac{y}{x}\right) = 0 \\ \Leftrightarrow P_{r,r}\left(\frac{1-x}{1-y}\right) = 0 \Leftrightarrow P_{r,r}\left(\frac{y}{x}\frac{1-x}{1-y}\right) = 0 \end{aligned} \tag{39}$$

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Finally, let R be the set of nonzero roots of $P_{r,r}$, and let c be a non-root, namely such that $P_{r,r}(c) \neq 0$: such a nonroot c definitely exists, as the degree of $P_{r,r}$ is at most $r < q$, hence not every element of the field can be a root. Considering the sets R and $c^{-1}R$, there are two possibilities:

- If $R \cap c^{-1}R = \emptyset$, it follows that $|R \cup c^{-1}R| = 2|R| \leq q-1 \Leftrightarrow |R| \leq \frac{q-1}{2}$.
- If $R \cap c^{-1}R \neq \emptyset$, assuming that $x \in R \cap c^{-1}R$, namely that both x and $y = cx$ are roots, it follows that $\frac{y}{x} = c$ is not a root, hence, by (39), neither are $c\frac{1-x}{1-cx}$ and $\frac{1-x}{1-cx}$: in other words, $\frac{1-x}{1-cx}$ does not lie in $R \cup c^{-1}R$. However, the map $x \to \frac{1-x}{1-cx}, x \in R \cap c^{-1}R$ is bijective. So, if we denote its range by U, we get that $(R \cup c^{-1}R) \cap U = \emptyset$, hence that $|(R \cup c^{-1}R) \cup U| = |R \cup c^{-1}R| + |U| = |R \cup c^{-1}R| + |R \cap c^{-1}R| = |R| + |c^{-1}R| = 2|R| \leq q-1$, hence $|R| \leq \frac{q-1}{2}$.

Thus, in both cases, we have reached the same bound for $|R|$. Adding 0 back into the set of roots leads to:

Theorem 4: (J. Sheekey): Let q be an odd prime power, and let $r \in [q-1] \backslash \{1\}$ be such that $(r, q-1) = 1$. Then, the number of roots of the polynomial $P_{r,r}(x) = (1-x)^r + x^r - 1$ is bounded above by $\frac{q+1}{2}$.

VIII. Summary and Conclusion

We studied the maximal cross-correlation of algebraically constructed Costas arrays (exponential Welch and Golomb arrays), and we discovered the following:

- that, surprisingly, the two constructions appear to be linked, despite being so different, as, for all primes except safe primes and 19, the maximal cross-correlations in the two cases differ systematically by 1 (see Theorem 3 and Conjecture 1); and
- that the maximal cross-correlation in both cases obeys the simple rule stated in Theorems 1 and 3, again with the exception of safe prime powers, 16, and 19.

All results presented in this work were inspired by the experiments synopsized in Tables I and II: for most prime powers a clear pattern of behavior emerged, which we were partly able to explain rigorously, while the missing pieces of the puzzle were formulated and left as conjectures for the time being. The "exceptional" prime powers that fail to conform to this pattern, consisting of the safe prime powers, 16, 19, and powers of 2 corresponding to Mersenne primes, defy even the formulation of a pattern of their own, let alone a conjecture. The only visible, but vague, pattern in their case is that, at safe prime powers and at powers of 2 corresponding to Mersenne primes, the maximal cross-correlation exhibits local minima for both the Golomb and the Welch cases. Therefore, they represent the most attractive families of frequency hopping patterns optimized for minimal autocorrelation, but, at the same time, resistant to cross-interference with other patterns in the family: they are the most suitable Costas arrays to use in a multiuser setting.

The challenges then for the immediate future are providing rigorous proofs for the (remaining) conjectures stated above on the one hand, and deciphering the behavior of the exceptional prime powers on the other.

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